Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/29/2020
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	miRNA Signature Detection and Countermeasures Against HZE Radiation Exposure for Tissue Degeneration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	08/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2021 per TRISH Task Book submission (Ed., 7/2/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Blaber, Elizabeth Ph.D. (Rensselaer Polytechnic Institute) Costes, Sylvain Ph.D. (NASA Ames Research Center) Grabham, Peter Ph.D. (Columbia University)		
Grant/Contract No.:	NNX16AO69A-T0404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will investigate the hypothesis that circulating microRNA (miRNA) signatures are driving microvascular disease and muscle degeneration associated with accelerating aging and will be enhanced by exposure to the space environment both in vivo (mice) and in vitro and test novel antagonist therapies to these miRNA signatures as countermeasures to reduce space radiation-induced health risks. A comprehensive Systems Biology approach will be used. Galactic cosmic rays (GCR) and solar particle events (SPE) scenarios will be used. Lastly, antagonists to the miRNA signatures will be synthesized using novel self-delivery miRNA technology and administered to the murine model with space irradiation. This will allow the possible development of a novel minimally invasive miRNA based radioprotector to be used as a countermeasure for space radiation. Collectively, understanding of how whole body space radiation impacts microvascular and tissue degeneration through circulating miRNAs will greatly enhance health risk prognostication and provide possible new mechanisms for protection against space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	IMPACT: Collectively, the work being done will provide strong evidence for biologically measurable molecular signatures in delineating health risk caused by the space environment through a novel systems biology approach. The generated data will allow novel mechanistic understanding of how circulating miRNAs are the essential regulators that control health risk due to HZE (High-Ionizing High Energy Particles) irradiation as well as form the basis of International Space Station (ISS) experiments and eventually lead to potential novel radioprotectors that can be utilized for disease prevention in astronauts. INNOVATION: The current work is using an innovative systems biology approach to study how a predicted circulating miRNA signature will impact health risk due to space irradiation. We utilized simulated microgravity conditions during and after the space irradiation to fully simulate conditions during spaceflight. Overall, there is limited knowledge in the NASA community concerning the role of miRNAs on tissues and even less is known regarding circulating miRNAs impact on the host at a system level. Additionally, overall health risk due to the combined effects of microgravity and space radiation are not understood well. We are pursuing the biological impact of the circulating miRNA signature associated with microgravity and space irradiation conditions through cutting edge countermeasure techniques. Through the use of a self-deliverable antagomir to target the miRNAs associated with space radiation and cardiovascular risk we are starting to show a minimally invasive technique to mitigate space radiation damage. This work allows the creation of a possible countermeasure to protect against long-duration space related biological damage.
Task Progress:	HYPOTHESIS and SPECIFIC AIMS: We hypothesize that there are critical circulating microRNA (miRNA) signatures in the blood that impact microvascular disease and muscle degeneration after exposure to whole body space irradiation and simulated microgravity (SMG). Additionally, antagonist therapies to these miRNA signatures have the potential to be used as countermeasures against space radiation. We will address this hypothesis by the following steps: 1) Specifically determine which miRNAs are secreted and circulating in the serum that impact microvascular disease and muscle degeneration due to spaceflight using in vivo models; 2) Test how these miRNAs will functionally and mechanistically impact vascular physiology using an in vitro model; 3) Use miRNA antagonists to knockout these miRNAs and determine if they mitigate the effects of the space environment in the in vitro model; and 4) Use the same miRNA antagonists in our in vivo models to develop a minimally invasive miRNA countermeasure. These aims are feasible with our significant experience and success with the requisite models in uncovering systemic host responses due to space radiation effects. SPECIFIC AIM 1: To determine the impact and mechanisms of circulating miRNA signatures that drive microvascular disease and muscle degeneration associated with aging with and without space irradiation and simulated microgravity. We have recently published on circulating (serum) miRNA signatures that can be used to determine both microgravity impact and space irradiation health risks. These findings will be expanded in the context of microvascular disease and muscle degeneration with space irradiation in combination with simulated microgravity. We will utilize miRNA-sequencing and droplet digital PCR (ddPCR) global profiling of circulating miRNAs to determine a conserved miRNA signature in the blood, liver, and related tissues (i.e., muscle and heart) after exposure to SMG and Galactic Cosmic Rays (GCR). For SMG we will use the hindlimb unloading model in vivo. For space radiation, mice will be irradiated using a consensus formula of 7 different ions to represent a total dose of 0.5 Gy GCR referred to as the simplified GCR sim model. We will also irradiate with an acute dose of 1 Gy of protons (50 MeV) to simulate Solar Particle Events (SPE). Blood from the mice will be collected before and after space irradiation and SMG to track circulating miRNA changes over time. Utilizing a space radiation disease risk model, the impact of the miRNAs will be used to predict the risk of microvascular disease and muscle degeneration due to space travel. SPECIFIC AIM 2: Establish functional significance and develop countermeasure strategies for circulating miRNAs and signaling pathways associated with microvascular disease and muscle degeneration with space irradiation and simulated microgravity (SMG). We hypothesize that the circulating miRNA signature will functionally impact cells to drive a more aggressive phenotype for microvascular disease and muscular damage when exposed to space radiation and SMG. Key miRNAs determined through preliminary data and from previous aims will be inhibited utilizing established CRISPR/Cas9 techniques and overexpressed (via viral techniques) in a 3D microvascular tissue model with and without HZE irradiation or SMG (using a custom made 3-D clinostat). Functional in vitro assays will be conducted using this model to determine the impact of the miRNA signature on key vascular biological processes associated with pathologies. Antagonists to the key circulating miRNA signature will allow for both a mechanistic understanding and the development of novel radioprotectors for health risks due to exposure to the space environment. Synthesized self-delivery miRNA antagonists will be applied to both the 3D tissue model and then the hindlimb unloading murine model with SMG and the radiation doses stated above. The tissues from the mice will be analyzed using ddPCR for quantification of miRNA and key pathways and genes will be assessed. Results from this aim will be used to enhance the previously established mathematical radiation health risk model. KEY FINDINGS:

	• We have determined a miRNA signature that is associated with GCR radiation • We have also determined a miRNA signature that is associated with SPE irradiation. • The above findings were done by miRNA-sequencing on mice with and without hindlimb unloading (HU) conditions (to simulated microgravity) irradiated with 0.5 Gy simplified GCR Sim, 1 Gy SPE sim, 5 Gy gamma, and unirradiated control mice. The following tissues were sequenced: liver, heart, serum, and soleus muscle. • Started countermeasure experiments with 3D microvasculature tissue model. We have started to show that by inhibiting three of the miRNAs in this GCR signature, we can mitigate the damage caused by GCR radiation. • Technology Requirements: The countermeasure development is occurring antagonists to the miRNAs. We have started a collaboration with Dr. David Kaplan to utilize his novel silk drug delivery system to package the antagomirs. This is system can act as a stable release of the antagomirs that can be utilized for long-term space missions. PROPOSED RESEARCH PLAN FOR YEAR 2: • Test the countermeasures with and without irradiation in mice under the HU conditions. • Further test the functional impact of the miRNA signature in the 3D microvasculature tissue model. • Finalize the countermeasure technology development.
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Articles in Peer-reviewed Journals	Schimmoller BJ, Trovão NS, Isbell M, Goel C, Heck BF, Archer TC, Cardinal KD, Naik NB, Dutta S, Rohr Daniel A, Beheshti A. "COVID-19 Exposure Assessment Tool (CEAT): Exposure quantification based on ventilation, infection prevalence, group characteristics, and behavior." Sci Adv. 2022 Sep 30;8(39):eabq0593. http://dx.doi.org/10.1126/sciadv.abq0593 ; PMID: 36179034; PMCID: PMC9524836 , Sep-2022